

RAHUL RAVI VK

College Park, MD | ravivk.rahul@gmail.com | 240-714-7128 | linkedin.com/in/RahulRaviVk | ravivkrahul.github.io/rahul-portfolio

SUMMARY

Robotics graduate student (UMD, M.Eng., GPA 3.9) with 6+ years of hands-on experience in industrial automation, process control, and instrumentation in safety-critical environments. Proven expertise in Honeywell DCS architecture, Siemens PLC troubleshooting, PID tuning, and SCADA systems — now applied to modern robotics through ROS 2, autonomous navigation, kinematic modeling, and MATLAB-based optimal control. Uniquely positioned at the intersection of real-world controls engineering and cutting-edge robotics research.

EDUCATION

M.Eng. in Robotics | University of Maryland, College Park Sep 2025 – May 2027
GPA: 3.9/4.0

B.Eng. in Instrumentation & Control | L.D. College of Engineering, Gujarat, India Aug 2014 – May 2018
GPA: 3.3/4.0

WORK EXPERIENCE

Assistant Manager, Instrumentation / KRIBHCO Fertilizers Ltd, Gujarat, India May 2023 – Jul 2025

- Led reliability initiatives for safety-critical process units, achieving **50% reduction in equipment downtime** through predictive maintenance and advanced control diagnostics.
- Integrated torque-converter speed regulation logic in **Honeywell DCS (C300)**, improving closed-loop stability and increasing plant availability by **15%**.
- Supervised deployment and validation of control logic updates during plant shutdowns, ensuring deterministic restart and minimal transient instability.
- Optimized instrumentation inventory forecasting, reducing carrying costs by **40%** while maintaining system reliability.

Senior Engineer, Instrumentation / KRIBHCO Fertilizers Ltd, Gujarat, India Oct 2019 – Apr 2023

- Designed **Siemens PLC** ladder logic to emulate bistable relay behavior, reducing false trips in Air Dryer Unit by **90%**.
- Performed closed-loop **PID tuning** on pressure and flow control loops, reducing overshoot and settling time, resulting in **30% decrease in process downtime**.
- Conducted root-cause analysis on oscillatory loop behavior and actuator nonlinearities to improve loop robustness.
- Commissioned full Air Dryer automation lifecycle: PLC sequencing, interlocks, alarms, field wiring, loop checking, and operational handover.

PROJECTS

Mobile Robot CAD, Simulation & ROS 2 Control

- Designed SolidWorks multi-link robot, exported to URDF and FBX (Blender); implemented autonomous point-to-point navigation via Python Proportional Controller in ROS 2/Gazebo. Integrated LiDAR and built a high-fidelity Digital Twin in Falcon Editor.

Optimal Controller Design for Mobile Robot – MATLAB

- Engineered GA-based trajectory generation (MATLAB Global Optimization Toolbox) and Gradient Descent-tuned Adaptive PID for dynamic tracking; validated closed-loop stability via Lyapunov analysis.

UR5 Kinematic Replication via Web Interface – ROS 2

- Built decoupled control architecture for UR5 manipulator in ROS 2, bridging a web interface to a Python backend via WebSockets to execute real-time inverse kinematics (ikpy) in Gazebo simulation.

Autonomous Maze Exploration – ROS 2 (C++17)

- Implemented ROS 2 Action Server encapsulating a DFS algorithm for asynchronous navigation with dynamic replanning upon detection of unexpected obstacles.

TECHNICAL SKILLS

- Control Systems:** PID, Adaptive PID, LQR, LQG, Kalman Filtering, State-Space Modeling, Lyapunov Stability, Trajectory Optimization, Closed-Loop Stability Analysis
- Industrial Automation:** Siemens PLC (Ladder Logic), Honeywell DCS (C300, Experion PKS), SCADA, Safety Interlocks, Control Valve Tuning, Loop Checking
- Robotics & Simulation:** ROS 2 (Humble, Actions/Services/Topics), Gazebo, MoveIt, URDF/Xacro, Digital Twin Simulation, Falcon Editor
- Programming:** C++17, Python (NumPy, SciPy, OpenCV, ikpy), MATLAB, WebSockets
- Tools & Hardware:** SolidWorks, Blender, Linux, Git, Raspberry Pi

CERTIFICATIONS

Siemens – Basics of PLC & SCADA | Honeywell – Experion PKS C300 Fundamentals